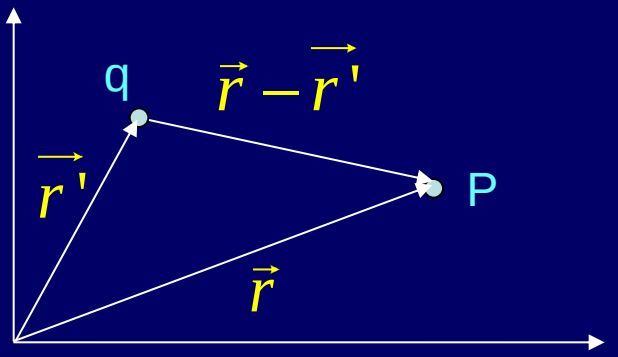
Electrostatic field, potential and work done

Recap: Single Charge

The electric field at a point P($\vec{r}$) due to a point charge q at ($\vec{r}\,'$) is,

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{\left|\vec{r}-\vec{r}\,'\right|^3}\left(\vec{r}-\vec{r}\,'\right)$$



Potential is derived from:

$$\vec{E}(\vec{r}) = -\vec{\nabla}\phi(\vec{r})$$

$$\phi(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{q}{\left|\vec{r}-\vec{r}\,'\right|} + C$$

Thus, the electrostatic potential is a scalar function of position whose negative gradient gives the electric field

Also can be defined as the work done in bringing a unit positive change from infinity to that point

Electrostatic field, potential and work done

Assembly of discrete charges

The electric field at a point P($\vec{r}$) due to a point charges q_i at (r_i') is,

$$\vec{E} = \sum_{i=1}^{n} \vec{E}_i = \frac{1}{4\pi\varepsilon_0} \left(\sum_{i=1}^{n} \frac{q_i}{\left| \vec{r} - \vec{r_i'} \right|^3} \left(\vec{r} - \vec{r_i'} \right) \right)$$

Potential:

$$\phi(\vec{r}) = \sum_{i=1}^{n} \frac{1}{4\pi\varepsilon_0} \frac{q_i}{\left| \vec{r} - \vec{r}_i' \right|} + C$$

Work done

$$W = \frac{1}{2} \sum_{i} \sum_{j<i} q_i \phi_j = \frac{1}{4\pi\varepsilon_0} \frac{1}{2} \sum_{i=1}^{N} \sum_{\substack{j=1 \\ j \neq i}}^{N} \frac{q_i q_j}{\left| \vec{r}_i - \vec{r}_j \right|}$$

Electrostatic field, potential and work done

Continuous charge distribution

Electric field:

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\int_C \frac{\lambda(\vec{r}')dl'(\vec{r}-\vec{r}')}{\left|\vec{r}-\vec{r}'\right|^3}$$

Potential:

$$\phi(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\int_C \frac{\lambda dl'}{\left|\vec{r}-\vec{r}'\right|} + C$$

Work done:

$$W = \frac{1}{2}\int \phi\lambda dl$$

Line charge, λ

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\int_S \frac{\sigma(\vec{r}')da'\left(\vec{r}-\vec{r}'\right)}{\left|\vec{r}-\vec{r}'\right|^3}$$

$$\phi(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\int_A \frac{\sigma da'}{\left|\vec{r}-\vec{r}'\right|} + C$$

$$W = \frac{1}{2}\int \phi\sigma da$$

Surface charge, σ

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\int_V \frac{\rho(\vec{r}')dV'\left(\vec{r}-\vec{r}'\right)}{\left|\vec{r}-\vec{r}'\right|^3}$$

$$\phi(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\int_V \frac{\rho dV'}{\left|\vec{r}-\vec{r}'\right|} + C$$

$$W = \frac{1}{2}\int \phi\rho dV$$

Volume charge, ρ

Electrostatic field and potential

Advantages of potential formalism

$$\vec{E} = -\vec{\nabla} V$$

If you know V, you can easily get E – just take the gradient

But the electric field is a vector quantity and has three components, whereas the potential is a scalar

How one function contains information about three independent functions?

The answer is that the three components are not independent but are *interrelated* from the fact that

$$\vec{\nabla} \times \vec{E} = 0 \Rightarrow \frac{\partial E_x}{\partial y} = \frac{\partial E_y}{\partial x}; \frac{\partial E_y}{\partial z} = \frac{\partial E_z}{\partial y}; \frac{\partial E_x}{\partial z} = \frac{\partial E_z}{\partial x}$$

Thus, electric field is a special kind of vector

Electrostatic field and potential

Advantages of potential formalism

Again $\vec{E} = -\vec{\nabla} V$

What do the fundamental equations for E look like in terms of V ?

$$\nabla . \vec{E} = \frac{\rho}{\varepsilon_0} = \nabla . (-\nabla V) = -\nabla^2 V \qquad (1)$$

Thus, in a region of charge density ρ

$$\nabla^2 V = -\frac{\rho}{\varepsilon_0}$$

Poisson's equation

In a charge-free region

$$\nabla^2 V = 0$$

Laplace's equation

$$\nabla \, X \, \vec{E} = 0 = \nabla \, X \, (-\nabla V) \qquad (2)$$

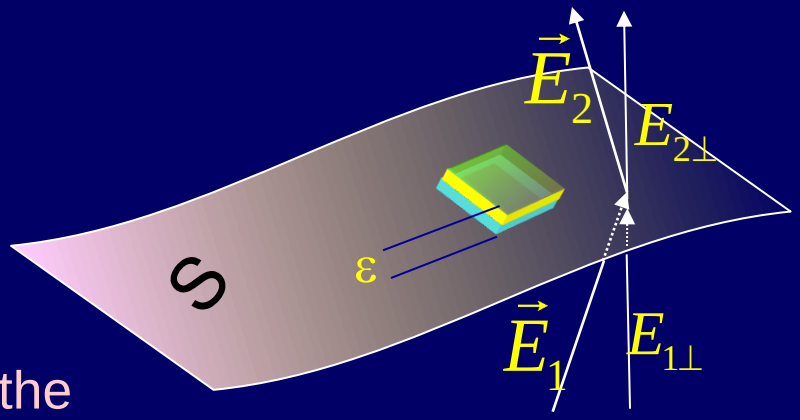
Curl of gradient is always zero, so it is fine

Electrostatic boundary conditions

When an electric field goes through a surface charge, it suffers a discontinuity

By how much is the discontinuity?

Consider a surface S of charges with charge density σ



The electric fields on either sides of the sheet and very close to the sheet, are

Resolve them in to components parallel and perpendicular to the surface S

Now, consider a small pill-box of infinitesimal thickness ε, forming a Gaussian surface over a small area δA of the surface S (the parts of the box above and below the surface S are shown separately).

From the Gauss' law, the flux of the electric field, through the box, is

$$\oint_{pillbox} \vec{E} \cdot \overrightarrow{da} = \frac{q_{encl}}{\varepsilon_0} = \frac{\sigma \delta A}{\varepsilon_0}$$

Electrostatic boundary conditions

In the limit of ε tending to zero, the flux through the thickness (sides) of the Gaussian surface can be neglected.

The electric field can be assumed to be uniform over δA. Then, flux becomes,

$$\left[E_{2\perp} - E_{1\perp}\right]\delta A = \frac{\sigma \delta A}{\varepsilon_0} \Rightarrow \left[E_{2\perp} - E_{1\perp}\right] = \frac{\sigma}{\varepsilon_0}$$

The normal component of the electric field is discontinuous by σ/ε_0

$$ie.\ it\ changes\ from - \frac{\sigma}{2\varepsilon_0}\hat{n}\ to\ \frac{\sigma}{2\varepsilon_0}\hat{n}\ at\ the\ boundary$$

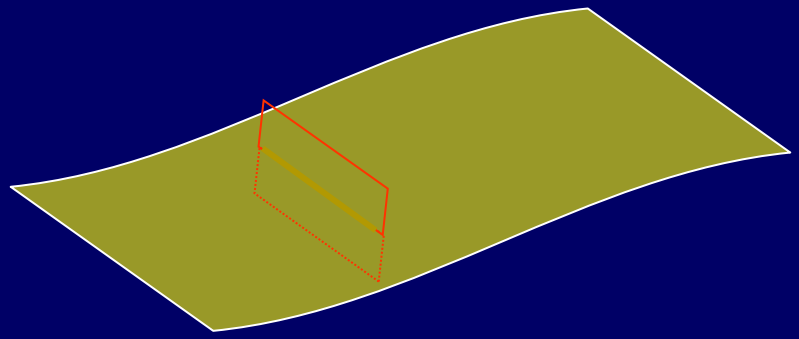
$\hat{n}$ is a unit vector perpendicular to the surface pointing upwards

When there is **no charge** on the surface, the normal component is continuous

Electrostatic boundary conditions

Let us now consider the components parallel (tangential) to the surface

We know that, $\oint_{\text{closed path}} \vec{E} \cdot \overrightarrow{dl} = 0$



Consider a loop of infinitesimal thickness ε, as shown

The tangential component of the electric field can be in any direction as long as it is parallel to the sheet. Let it be parallel to the length *l* of the loop. In the limit $\varepsilon \rightarrow 0$,

$$\oint_{\text{closed path}} \vec{E} \cdot \overrightarrow{dl} = E_{2||} l - E_{1||} l = 0 \qquad E_{2||} = E_{1||}$$

the tangential component is continuous

Electrostatic boundary conditions

The boundary conditions can be combined to obtain a single condition

$$\vec{E}_2 - \vec{E}_1 = \frac{\sigma}{\varepsilon_0}\hat{n}$$

What about the potential?

$$V(r_2) - V(r_1) = -\int_{r_1}^{r_2} \vec{E}\cdot d\vec{l} = 0$$

Thus the potential is continuous across the surface or any boundary
However, the gradient of the potential (ie. the electric field inherits discontinuity

Since $\vec{E} = -\vec{\nabla}V$

$$\vec{\nabla}V(r_2) - \vec{\nabla}V(r_1) = -\frac{\sigma}{\varepsilon_0}\hat{n}$$

Taking the dot products with vector $\hat{n}$

$$\vec{\nabla}V(r_2).\hat{n} - \vec{\nabla}V(r_1).\hat{n} = -\frac{\sigma}{\varepsilon_0}$$

$$\Rightarrow \frac{\partial V_2}{\partial n} - \frac{\partial V_1}{\partial n} = -\frac{\sigma}{\varepsilon_0}$$

Work and Energy

The work done in assembling discrete charges is given by

$$W = \frac{1}{2}\sum_{i}\sum_{j<i} q_i \phi_j = \frac{1}{4\pi\varepsilon_0}\frac{1}{2}\sum_{i=1}^{N}\sum_{\substack{j=1 \\ j\neq i}}^{N}\frac{q_i q_j}{\left|\vec{r}_i - \vec{r}_j\right|}$$

For Volume charge ρ

$$W = \frac{1}{2}\int \phi\rho dV$$

Using Gauss' law

$$\vec{\nabla}\cdot\vec{E} = \frac{1}{\varepsilon_0}\rho$$

$$W = \frac{\varepsilon_0}{2}\int \phi\vec{\nabla}.\vec{E}dV$$

Using now

$$\vec{\nabla}.(\phi\vec{E}) = \vec{\nabla}\phi.\vec{E} + \phi\vec{\nabla}.\vec{E}$$

$$W = \frac{\varepsilon_0}{2}\left[\oint \phi\vec{E}.d\vec{a} - \int\left(\vec{\nabla}\phi.\vec{E}\right)dV\right] = \frac{\varepsilon_0}{2}\left[\oint \phi\vec{E}.d\vec{a} + \int\left|\vec{E}\right|^2 dV\right]$$

Work and Energy

$$W = \frac{\varepsilon_0}{2}\left[\oint \phi \vec{E}.d\vec{a} + \int \left|\vec{E}\right|^2 dV\right]$$

Integral over volume containing all the charges

In the first term, Φ varies as 1/r and electric field varies as $1/r^2$ and area element varies as r^2, therefore the integrand varies as 1/r. Therefore when we integrate through larger and larger area, the first term will keep decreasing and the second term keeps increasing, keeping the total W constant. The first term reduces to zero when we integrate through entire space…in that condition

$$W = \frac{\varepsilon_0}{2} \int_{\text{all space}} \left|\vec{E}\right|^2 dV$$

It does not matter whatever volume over which you integrate (as long as it encloses all the charges), you get the correct W.

Further, this shows that energy stored in charges are always positive

Work and Energy

If so, the Energy of a positive and negative charge separated by a distance r is (the work done in bringing –q from infinity to r away from +q

$$W = -\frac{1}{4\pi\varepsilon_0}\frac{q^2}{r}$$

which is negative, ..so what's wrong here??

As per energy eqn., Energy stored in the electric field of a point charge

$$W = \frac{\varepsilon_0}{2}\frac{1}{\left(4\pi\varepsilon_0\right)^2}\int \frac{q^2}{r^4}\left(r^2 \sin\theta dr d\theta d\varphi\right) = \frac{q^2}{8\pi\varepsilon_0}\int_0^\infty \frac{1}{r^2}dr = \infty$$

Again, what's wrong here. Which equation one must use??

Important: Equation for discrete charges does not consider creating the charges itself, whereas the work done (energy stored) obtained by electric field does consider all contribution. In continuous distribution amount of charge at a point is vanishingly small and it's contribution to the potential is zero.

Work and Energy

Where is the energy stored ?: in the charges or in the field??
See the two equations for W – integration over (1) field or (2) charge

$$W = \frac{\varepsilon_0}{2} \int_{\text{all space}} \left|\vec{E}\right|^2 dV \qquad W = \frac{1}{2}\int \phi\rho dV$$

For a spherical shell, the charge is confined to the surface, but the field is present everywhere outside this surface

In the radiation theory, the energy is considered to be stored in field and energy density is given as

$$\frac{\varepsilon_0}{2}\left|\vec{E}\right|^2 = \text{Energy per unit volume}$$

In the electrostatics, one can as well say the energy is stored in charges with density

$$\frac{\rho\phi}{2} = \text{Energy per unit volume}$$

Work and Energy

Superposition of energy possible ???

$$W = \frac{\varepsilon_0}{2} \int_{\text{all space}} \left|\vec{E}\right|^2 dV$$

For two fields

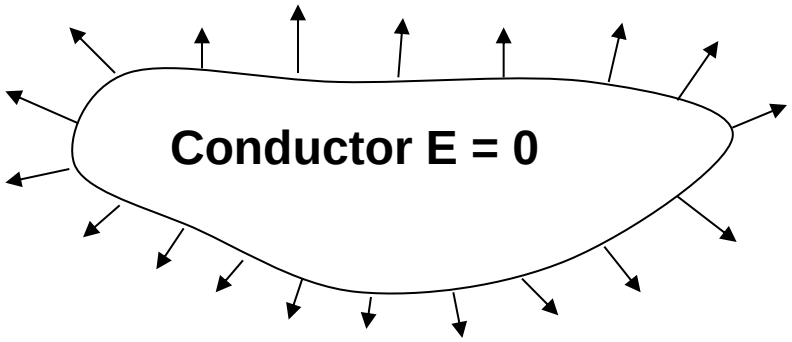
$$W_{total} = \frac{\varepsilon_0}{2} \int_{\text{all space}} (\vec{E}_1 + \vec{E}_2)^2 dV = \frac{\varepsilon_0}{2} \int_{\text{all space}} (E_1^2 + E_2^2 + 2\vec{E}_1.\vec{E}_2) dV$$

$$W_{total} = W_1 + W_2 + \varepsilon_0 \int \vec{E}_1.\vec{E}_2 dV$$

Since electrostatic energy is quadratic in field, the compound system does not obey the superposition principle, there will be cross terms.

Conductors

Basic properties

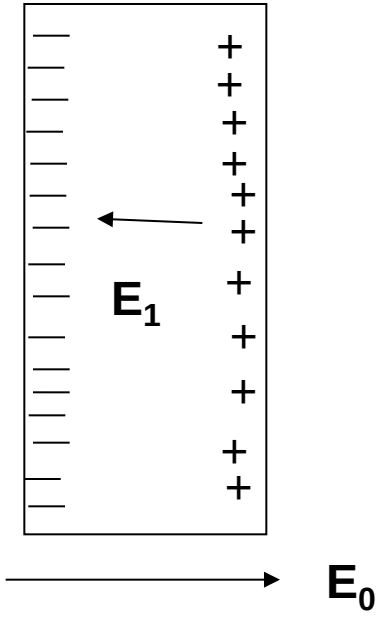


1) E = 0 inside a conductor

Why? If there were any field, the free charges would move, and it would not be electrostatics anymore….that's alone not the reason

If you put the conductor in an external field E_0 the free charges move in respective directions…. create induced charges, produce **E_1** opposite to **E_0**

The whole process is almost instantaneous



Conductors

Basic properties

2) $\rho = 0$ inside a conductor

Follows from Gauss's law $\vec{\nabla} \cdot \vec{E} = \frac{1}{\varepsilon_0} \rho$

If E =0, then $\rho = 0$

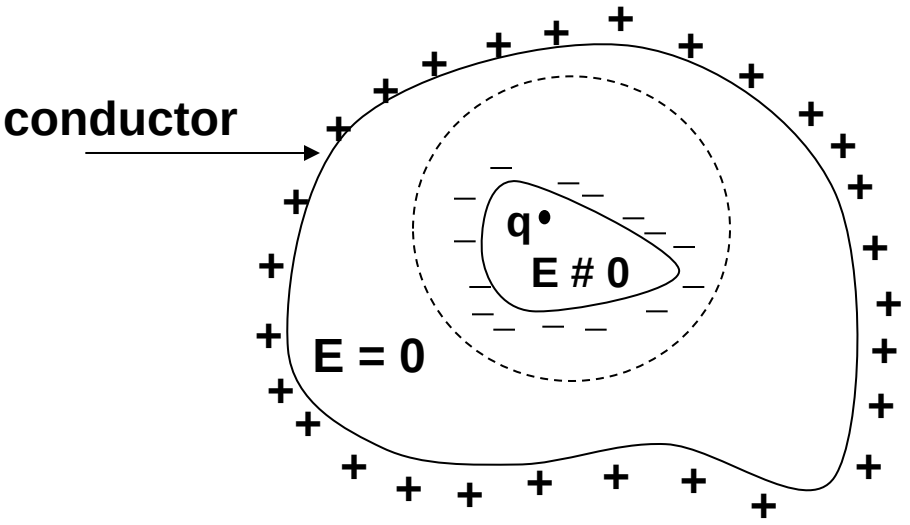
Charges still exist, but equal amounts of + and -

3) Any net charge reside on the surface (that is the only possibility)

4) A conductor is an equipotential surface

$$V(a) - V(b) = -\int_a^b \vec{E} \cdot d\vec{l} = 0$$ since E=0 hence $V(a) = V(b)$

5) E is perpendicular to the surface, just outside a conductor



Field inside a conductor is zero
But if there is a cavity inside the
Solid conductor, the field is not zero
Inside the cavity

Surface charge and the Force on a conductor

Field inside a conductor is zero

Boundary condition implies that field immediately out side

$$\vec{E} = \frac{\sigma}{\varepsilon_0}\hat{n}$$

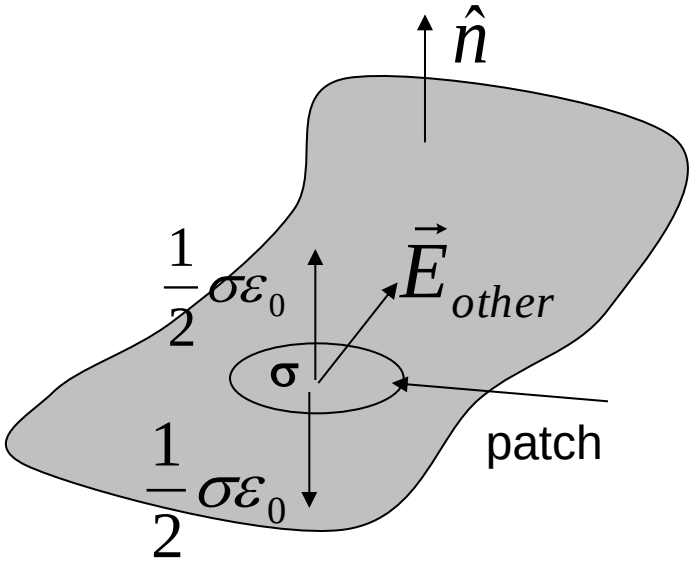
In the presence of an electric field, a surface charge will experience a force

Force per unit area is $\vec{f} = \sigma\vec{E}$

What is the $\vec{E}$ here????

Consider a tiny, small patch around the point under consideration. The total field at the patch is

$$\vec{E} = \vec{E}_{patch} + \vec{E}_{other}$$



The patch cannot exert any force on itself, it is like sitting in a basket and trying to lift it with the handles (alternate argument would be, equal force on either side due to equal fields)

Hence Force on the patch is entirely due to $\mathbf{E}_{\mathbf{other}}$

The discontinuity is entirely due the patch, which puts out a field $\sigma/2\varepsilon_0$ on either side pointing away from the surface

$$\vec{E}_{above} = \vec{E}_{other} + \frac{\sigma}{2\varepsilon_0}\hat{n} \qquad \vec{E}_{below} = \vec{E}_{other} - \frac{\sigma}{2\varepsilon_0}\hat{n}$$

$$\vec{E}_{other} = \frac{1}{2}\left(\vec{E}_{above} + \vec{E}_{below}\right) = \vec{E}_{average}$$

This argument applies to any surface charge. In the case of a conductor, field is zero inside and σ / ε_0 outside.

Hence, $E_{average}$ will be $\frac{\sigma}{2\varepsilon_0}\hat{n}$

Therefore $$\vec{f} = \sigma\vec{E} = \frac{1}{2\varepsilon_0}\sigma^2\hat{n}$$

This amounts to an outward electrostatic pressure on the surface

Capacitors

Since **E** is proportional to Q, so also V is proportional to Q

Capacitance is the constant of proportionality

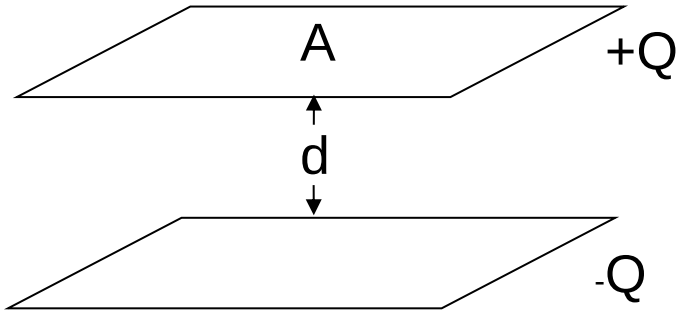
$$C = \frac{Q}{V}$$

Measured in farads: coulomb per volt

Capacitance is an intrinsically positive quantity

Example 1.

Find the capacitance of a parallel plate capacitor consisting of two metal surfaces of area A held a distance d apart.



Surface charge density σ = Q/A

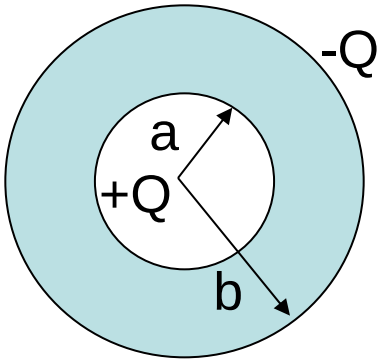
$$\left|\vec{E}\right| = \frac{\sigma}{\varepsilon_0} = \frac{1}{\varepsilon_0}\frac{Q}{A}$$

$$V = \frac{Q}{A\varepsilon_0}d$$

$$C = \frac{Q}{V} = \frac{\varepsilon_0 A}{d}$$

Example 2.

Capacitance of two concentric spherical shells



Field between the spheres is $\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\hat{e}_r$

$$V = -\int_b^a \vec{E}\cdot\vec{dl} = -\frac{Q}{4\pi\varepsilon_0}\int_b^a \frac{1}{r^2}dr = \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{a}-\frac{1}{b}\right)$$

$$C = \frac{Q}{V} = 4\pi\varepsilon_0\frac{ab}{b-a}$$

Work done to charge up a capacitor

Let q be the charge on the positive plate at some instant

Potential difference is q/C

Work that must be done to bring in the next piece of charge dq

$$dW = dqV = \frac{q}{C}dq$$

Total work necessary then to go from q = 0 to q = Q

$$W = \int_0^Q \left(\frac{q}{C} \right) dq = \frac{1}{2} \frac{Q^2}{C}$$

Since Q = CV

$$W = \frac{1}{2} CV^2$$

V is the potential across the capacitor